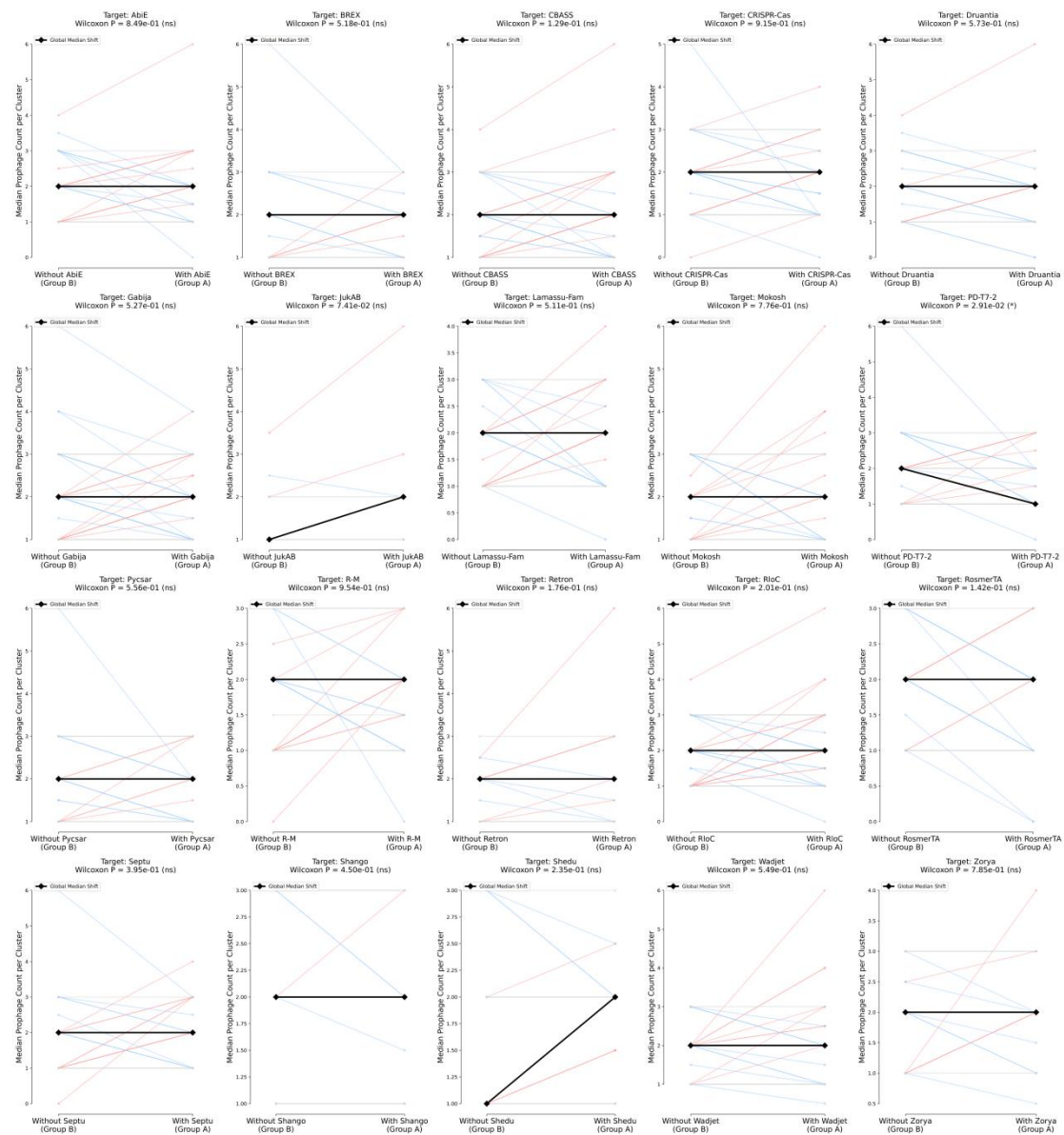
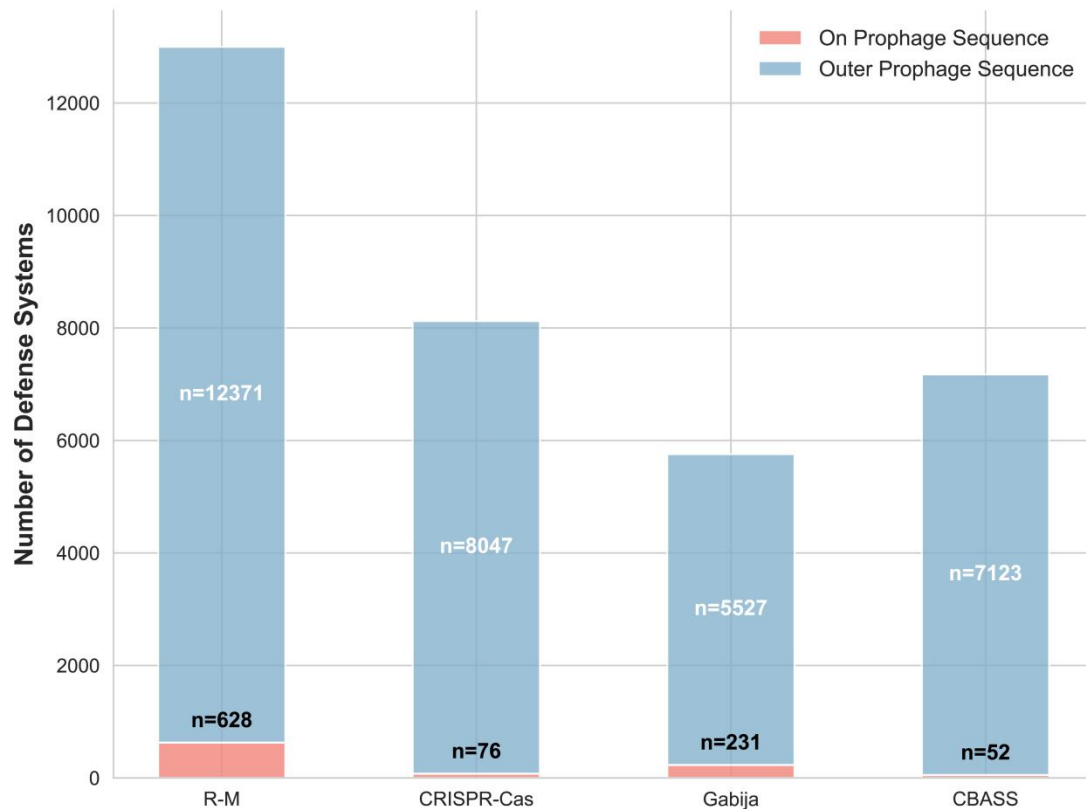




Supplementary Fig 2: Prophage abundance across the 20 most prevalent defense systems. Prophage counts are compared between genomes encoding a specific defense system (Group A) and those lacking it (Group B) within each Mash group. Colored lines denote individual Mash groups: red lines indicate higher prophage abundance in Group A versus Group B, while blue lines denote lower abundance. Black lines indicate the global median trend across all groups. Statistical significance was assessed using the Wilcoxon rank-sum test, with P -values indicated above each panel.



Supplementary Fig 3: Distribution of defense systems detected within prophage regions and their relationship with genomic defense system abundance.

Stacked bar plots show the number of representative defense systems identified within prophage-associated regions. The blue portion indicates defense system elements located in sequences outside prophage regions (outer prophage sequences), whereas the red portion represents defense system elements directly embedded within prophage sequences (on prophage sequences).